

8 Interrogative clauses

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So far, we have developed a semantic system that compositionally derives semantic values for “declarative” clauses, whether standing on their own as main or matrix sentences or embedded as constituent clauses (relative clauses, complement clauses, adjunct clauses). Extensionally speaking, declaratives are of type t and their denotation is determined relative to an evaluation index (a world, or a world-time pair) and a variable assignment: $\llbracket \phi \rrbracket^{i,g}$. The system also supplies (possibly relative to a variable assignment) an intension of type $\langle s, t \rangle$: a proposition, or a function from indices to truth-values, $\llbracket \phi \rrbracket_e^g = \lambda i. \llbracket \phi \rrbracket^{i,g}$.

In this chapter, we will expand the system to deal with a central kind of non-declarative clause: interrogatives. These also can stand on their own:

- (1) a. Did Sakina see Emily?
- b. Did Sakina see Emily or did Sakina see Julie?
- c. Who did Sakina see?

In set talk: a set of indices. In the simplest system: a set of worlds.

Another important kind of non-declarative are imperatives, if you're interested in which, you should consult Portner 2007, M. Kaufmann 2012, von Stechow & Iatridou 2017.

And they can appear as embedded clauses:

- (2) a. Melanie wonders whether Sakina saw Emily.
 b. Inessa is surprised who Sakina saw.
 c. Whether Sakina saw Emily or not, she definitely saw Julie.

The semanticist's task in the analysis of interrogative clauses is to propose suitable types of semantic values for them and to show how these semantic values are built up compositionally. But before we apply ourselves to this task, it is useful to say a little bit about speech acts, and about the relation between semantic values and speech acts.

8.1 *A little pragmatics*

Leaving semantics out of the picture for the moment, interrogative clauses differ from declarative clauses in their syntax and in their pragmatics. In English, interrogative clauses can be told apart syntactically from declarative clauses by the presence of a clause-initial *wh*-phrase and/or the inverted order of subject and auxiliary. The terminology “interrogative”/“declarative”, however, alludes to a distinction not in grammatical form but in communicative function. When uttered as main clauses (i.e., not embedded in a larger structure), declarative sentences typically serve to make assertions, whereas interrogative sentences serve to ask questions. These two clause-types serve to perform different kinds of speech acts.

Stalnaker 1978 outlined an influential formal model of what happens in a conversation and what it means to make an assertion. The central concept is that of a body of publicly shared information, or “common ground”, which evolves as the conversation proceeds. A proposition is in the common ground if each interlocutor is “disposed to act as if he assumes or believes the proposition is true, and as if he assumes or believes that his audience assumes or believes that it is true as well”. The common ground can be characterized by the set of worlds in which every proposition in the common ground is true; Stalnaker calls this set of worlds the “context set”. The act of making an assertion is a proposal to update — in fact, to shrink — the context set. The particular way in which the context set is to be shrunk depends on the semantic value of the asserted sentence, more specifically, on its intension. To assert a sentence ϕ is to propose that the current context set c be replaced by a new context set which is the intersection of c with the intension of ϕ , i.e., $c \cap \llbracket \phi \rrbracket_c$.

Against this backdrop, how might we think about the speech act of asking a question? What is the point of this speech act? Questions, unlike assertions, don't provide information about the world and therefore do not in themselves lead to updates of the common ground. Their purpose rather is to constrain the future course of the conversation in a certain way. In the absence of any particular question under consideration,

Not “shrink” in the sense of reducing cardinality. The cardinality of the context set may never become less than uncountably infinite.

We're leaving off the variable assignment parameter, which is legitimate if there are no free variables in ϕ .

At least this is true if we stick to questions that don't have presuppositions (or to questions whose presuppositions are already common ground at the point when they are asked). When a question that has a presupposition is asked in a context where the presupposition is not yet in the common ground, a listener can get new information by accommodating this presupposition.

a speaker might assert whatever they find interesting or useful. But if someone asks a question, they thereby express an expectation that the next conversational move will be one out of a very limited set of possible assertions—intuitively, the next assertion will “address the question” rather than provide some random other information.

Here is a way to formally model this intuition. A question partitions the context set in a certain way, and it amounts to a request for assertions which, so to speak, “carve up” the context set along the lines determined by this partition. The requested next assertion should not make any distinctions between worlds which the partition lumps together.

Here’s the formal definition of a partition:

Definition 1 (Partition) A set $\mathcal{P}$ is a *partition* of a set A iff

- (i) every element of $\mathcal{P}$ is a non-empty subset of A ,
- (ii) any two non-identical elements of $\mathcal{P}$ are disjoint from each other, and
- (iii) the union of all the elements of $\mathcal{P}$ is A .

The elements of a partition are also called its cells.

With this definition in hand, we refine Stalnaker’s model of conversation. Each stage in a conversation is now characterized not by its context set, but by its “partitioned context set”, i.e., by a partition of some set of possible worlds. The union of this partition corresponds to Stalnaker’s old context set, i.e., it contains the worlds compatible with every proposition in the common ground. The partitioning represents the interlocutors’ shared commitment not to make distinctions between worlds that are “cell-mates”—at least not for the time being. This is a renegotiable commitment and, as we will see, it only stays in force until someone raises a new question. The definition of “relevant assertion” in Definition 3 below spells out precisely what the commitment amounts to. In Definition 2, we adapt Stalnaker’s characterization of the context-changing role of assertions to our new, more elaborate set-up.

Definition 2 (Update by assertion) To assert a sentence ϕ is to propose that the current partitioned context set C be replaced by a new partitioned context set which is constructed by intersecting each cell of C with the intension of ϕ . More precisely, C is to be replaced by

$$\{p: p \neq \emptyset \ \& \ \exists p' \in C. p = p' \cap \llbracket \phi \rrbracket_e\}$$

Definition 3 (Relevance) An assertion is *relevant* w.r.t. to a partitioned context set C iff the update it proposes results in a subset of C .

The notion of “relevance” that is formalized here calls for a bit of clarification. First, here are a couple of more commonly used (and more transparent) definitions.

The idea of refining the Stalnakerian model of the context by moving from the context set to a partitioned context set goes back to Groenendijk 1999. A formally equivalent way is to model the context as an equivalence relation (see, for example, Jäger 1996 and Hulstijn 1997). We will later see approaches that give a partition semantics to interrogatives. It’s important to distinguish the two moves. As we’ll see in this section, one can have a partition pragmatics without a partition semantics. This theme is explored in much more sophisticated detail in Fox 2018.

Starr 2020 adds to the partitioned context set model a way of tracking the speech act effect of imperatives.

Our proposed context model is simple compared to some other well-known systems, such as Roberts 2012 and Farkas & Bruce 2010.

Definition 4 (Relevance') A proposition p (i.e. set of worlds) is relevant w.r.t. to a partition Q of a set of worlds iff p is identical to a cell in Q or to a union of cells in Q .

Definition 5 (Relevance'') A proposition p is relevant w.r.t. to a partition Q of a set of worlds iff there is no pair of worlds w and w' such that both

- (i) there is a cell $c \in Q$ such $w \in c$ & $w' \in c$
(i.e., w and w' are cell-mates),
- (ii) and $w \in p$ & $w' \in p$
(i.e. p has different truth values in w and w').

These two definitions differ slightly in that Definition 5 (unlike Definition 4) allows p to include arbitrary worlds that are not in any cell of Q at all. But both express the essential requirement that a relevant proposition must not discriminate among cell-mates. It must not “cut up” the space of worlds (in the union of the partition) in any way that does not coincide with the boundaries between cells.

Definitions 4 and 5 apply to a proposition, whereas our Definition 3 applies to an assertion and presupposes the rule for update-by-assertion. But apart from this difference, Definition 3 captures the same notion of relevance. (To make the connections fully precise: If C is a partitioned context set and ϕ is an asserted sentence, then the assertion of ϕ is “relevant” w.r.t. C in the sense of Definition 3 iff $\bigcup C \cap \llbracket \phi \rrbracket_c$ is “relevant” w.r.t. C in the sense of Definition 4, which holds, in turn, iff $\llbracket \phi \rrbracket_c$ is “relevant” w.r.t. C in the sense of Definition 5. Prove these equivalences as an exercise.)

To get an intuitive appreciation of this concept, it is important to see that “relevance” in this technical sense is a purely negative requirement, so to speak: it is just the absence of any irrelevant (i.e., not specifically asked for) information. By this criterion, a “relevant” answer may be totally uninformative; it could be a tautology, or any other proposition that is already entailed by the common ground. So evidently, relevance is not a sufficient condition for appropriate answers; there are additional conditions, notably that an answer — or any assertion, for that matter — should be informative. It is also worth pointing out that the formal requirement of relevance is *prima facie* violated by many answers which in actual fact are judged felicitous, e.g., when somebody responds to the question *Is it raining?* by saying: *Well, Becky just came in dripping wet*, or *Maybe*, or *I don't know*. To make sense of this wider range of intuitively felicitous answers, a proponent of the present formal model needs to say that in these cases, the answerer is implicitly changing or amending the question that was explicitly asked, and instead is addressing a different (though related) question that they don't formulate explicitly but expect their audience to infer (“accommodate”).

Now the only piece of our story that is missing is a recipe for “update by question”. The idea, as we have said, is that asking a question amounts to proposing a specific replacement of the current partitioned context set by a new one. As in the case of assertions, the construction of the new partitioned context set should be determined by the semantic value of the sentence that was uttered. We know what we have in mind for particular examples. E.g., when someone asks *Is it raining?*, the intended outcome is a two-membered partition, with one cell consisting of worlds in which it is raining and the other cell of worlds in which it is not raining. If the question is a so-called “alternative question” like *Is Julie in Barcelona or did Emily call them on the phone?*, the result should be a four-celled partition corresponding to the four possible configurations of truth-values for the two component sentences. If the question asked is *Who among these two (Carli and Jodie) came to the party?*, the result should be a partition with four cells: one with worlds in which Carli and Jodie both came, one with worlds in which Carli but not Jodie came, one with worlds in which Jodie but not Carli came, and one with worlds in which neither Carli nor Jodie came.

What is the general recipe by which these outcomes are obtained as a function of the semantic value of each interrogative sentence? To answer this question, we have to carry out the semanticist’s task, i.e., assign semantic values to interrogative clauses. We will do so in the next section, but for now we will articulate a basic intuition about the semantic value of interrogatives and explore how they can be used to construct a partitioned context set.

Perhaps the most widespread idea about the semantic value of interrogatives is that they denote *sets of propositions*. The particular structural features of interrogatives are taken to serve to both “lift” the type of the denotation from the type of propositions (the denotation of declaratives) to the type of sets of propositions, and to generate the individual propositions in the set. We will look at the mechanics of this in the next section. For now, let’s assume the following semantic value for the sample *wh*-questions *Who among these two (Carli and Jodie) came to the party?*:

- (3) $\llbracket \text{Who among these two (Carli and Jodie) came to the party?} \rrbracket = \{\lambda w. \text{Carli came to the party in } w, \lambda w. \text{Jodie came to the party in } w\}$

Now, notice that the propositions in this set are not disjoint (they can both be true of the same world) and that their union leaves out the worlds where neither Carli nor Jodie came to the party. So, this set of propositions is not a partition of the set of all possible worlds and also not a partition of any context set with minimal information about who came to the party. So, there’s apparently not a seamless link between our partition pragmatics for question acts and our semantic values for interrogatives.

Should we therefore redo our semantics so that it does, after all, map interrogative clauses to semantic values that are partitions? This can be done and has been argued for, and we will return to a discussion of this possibility. For now, however, we observe that it is not necessary. An equally reasonable take on the discrepancy between our semantics and our pragmatics is that it simply reflects the division of labor between semantics and pragmatics. We don't necessarily need a semantics that directly delivers partitions as semantic values. All we need is to make precise how the semantic values delivered by the semantics help determine the partitions that figure in the pragmatics. In other words, we need to formulate our pragmatic rule for "update by question" in such a way that it spells out how the partition effected by a question-act is determined by the semantic value of the uttered sentence. We will work up to this formulation in a few steps. As we will see, there is a straightforward and general recipe for converting an arbitrary set of propositions into a partition of a given set of worlds, and our update principle will make use of that recipe.

Informally speaking, each proposition in the question-denotation defines a "dividing line" across the space of worlds in the context set, and as we use multiple propositions to draw multiple lines across this space, we get increasingly fine-grained partitions as the number of propositions goes up. The first proposition cuts the space into two cells (the region where it is true and the region where it is false). The second proposition subdivides these two cells further into four; the third proposition leaves us with eight cells; and so on.

Another way of describing this process is that each new proposition introduces a new condition that cell-mates must satisfy. At the beginning, before any line has been drawn, all of the context set is one big cell and every world counts as a cell-mate of every other one. Then we draw a line for the first proposition (call it p_1), and if two worlds differ with respect to the truth-value of p_1 , they now are no longer cell-mates. w and w' are cell-mates now only if $p_1(w) = p_1(w')$. With the second proposition, p_2 , we draw a second line and "break up" yet more of the original cell-mate relations. At this point, only those pairs of worlds remain cell-mates which are treated alike by both p_1 and p_2 . I.e., w and w' are cell-mates now iff $p_1(w) = p_1(w') \ \& \ p_2(w) = p_2(w')$. And so on for the third and all other propositions in the given set of propositions.

So a set of propositions can be used to define a "cell-mate relation", and thus a partition.

This assumes that the propositions are all "logically independent" of each other and of the propositions in the original common ground. That is, none of them contradicts or is entailed by any other (or any conjunction or disjunction of others). If there are logical relations between the propositions, then the total number of cells is smaller. Note that the definition of "partition" requires all cells to be non-empty, so the intersection of two incompatible propositions is not a cell.

- (4) Let $\mathcal{P}$ be a set of propositions and c a set of worlds.
- a. $\sim_{\mathcal{P},c}$, the cellmate relation in c based on $\mathcal{P}$, is defined as follows:
 $\forall w, w': w \sim_{\mathcal{P},c} w' \text{ iff } w \in c \ \& \ w' \in c \ \& \ \forall p \in \mathcal{P}: p(w) = p(w').$
 - b. $\text{PARTS}(\mathcal{P}, c)$, the partition of c based on $\mathcal{P}$, is defined as follows:
 $\text{PARTS}(\mathcal{P}, c) = \{p: \exists w \in c. p = \{w': w \sim_{\mathcal{P},c} w'\}\}$

The cellmate-relation is an equivalence relation (reflexive, symmetric, and transitive). It holds between any two worlds that give the same truth-value to each proposition in the “seed set”. The corresponding partition consists of the subsets of c that are made up of worlds that are cell-mates.

We are ready to propose a preliminary rule of “Update by question”:

Definition 6 (Update by question (draft)) To ask a question by uttering a sentence ϕ that denotes a set of propositions $\mathcal{P}$ is to propose that the current partitioned context set C be replaced by the new partitioned context set $\text{PART}(\mathcal{P}, \cup C)$.

In the next section, we turn to the task of compositionally deriving the semantic value of an interrogative.

8.2 Compositional computation of semantic values for interrogative clauses

8.2.1 An updated version of Karttunen 1977

Interrogative clauses are built largely with the same lexicon and syntactic rules as declarative clauses; but they also contain certain interrogative-specific morphemes, syntactic features, or functional heads that make crucial contributions to their non-declarative meanings. Following Karttunen 1977, our syntax for English posits an abstract (i.e., silent) complementizer that has a non-vacuous semantics, and a feature on *wh*-phrases that is semantically vacuous but subject to a certain distributional constraint.

- (5) Karttunen’s “proto-question” operator, syntactically a C-head:
 $\llbracket ? \rrbracket^w = \lambda p_{st}. \lambda q_{st}. p = q.$
- (6) lexical entries for interrogative words, e.g.:
 $\llbracket \text{who}^{[wh]} \rrbracket^w = \lambda f_{et}. \exists x [x \text{ is human in } w \ \& \ f(x) = 1]$

The meaning of *who* is exactly the same as the meaning of *somebody*. We will address the role of the feature $[wh]$ in Section 8.2.3.

In the syntactic derivation of a constituent question, *wh*-movement applies and puts the *wh*-word above the operator in C. For the example *Who did Sakina see?* or *who Sakina saw* (we will ignore tense and auxiliary *do*, and thereby ignore the difference between matrix and embedded versions), this gives rise to a structure like (7):

Note that this update rule implies that each new question “wipes out” the current partitioning of the context set and replaces it by a new one. This may not be right and isn’t the only way we can go. We may want a system where successive questions can serve to *refine* the partition. Even more complex possibilities exist, as for example in Roberts 2012.

The implementation here differs from Karttunen 1977 in not using the Montague Grammar framework; instead we presuppose a division of labor between syntax and semantics in line with Heim & Kratzer 1998. Aside from implementation, there are some small substantive differences as well. Most saliently, the extensions we will assign to interrogative clauses will include both true and false propositions. (In Hagstrom 2003’s terminology, they will correspond to ANS-POSS rather than ANSTRUE.) Moreover, we will depart from Karttunen in the treatment of polar questions.

Karttunen’s motivation for including a restriction to true propositions had to do with the semantics of question-embedding constructions. So the time to discuss it is when we get to embedded questions. Right now we focus on matrix questions.

The official denotation in (5) maps propositions to functions from propositions to truth-values, or equivalently, it maps propositions to characteristic functions of sets of propositions. If we replaced these characteristic functions by the corresponding sets, (5) would read as follows:

$$(5') \quad \llbracket ? \rrbracket^w = \lambda p. \{p\}$$

In other words, it would be a function that maps each proposition to the singleton set that has it as its only member.

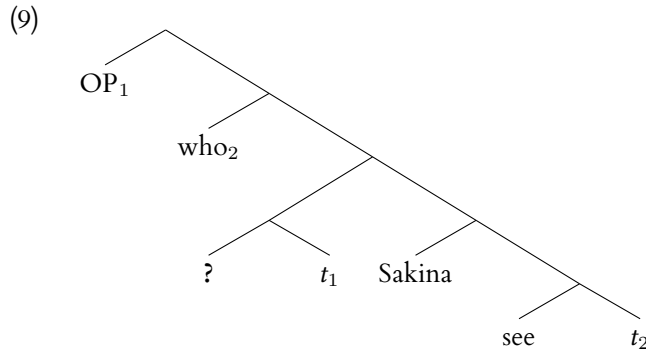
(7) who 2 [? Sakina see t_2]

Unfortunately, when we examine the semantic types at each node in (7), we discover a type-mismatch that prevents us from interpreting the top-most node. We fix this problem by positing a slightly more complex C-head: We base-generate ? together with another, covert operator as its sister. This covert operator is semantically vacuous, but it can move and leave a trace of type $\langle s, t \rangle$. The full syntactic derivation for a constituent question then proceeds as in (8).

(8) Who did Sakina see?

[_C ? OP] Sakina see who
 who 2 [[? OP] Sakina see t_{2e}] (wh-movement)
 OP 1 [who 2 [[? t_{1st}] Sakina see t_{2e}]] (operator movement)

The resulting LF looks thus like this in tree format:



We will usually drop the type-labels on the traces, but keep in mind that the operator leaves a trace of type $\langle s, t \rangle$ (i.e., a variable over propositions, not individuals). Accordingly the topmost application of Predicate Abstraction will yield a function from propositions to truth-values (type $\langle st, t \rangle$, the characteristic function of a set of propositions). Let's compute the meaning of this LF:

(10) Computation for LF (9):

$\llbracket 1. \text{ who } 2. [? t_1] \text{ Sakina see } t_2 \rrbracket^{w, \emptyset}$
 = (by Predicate Abstraction)
 $\lambda p. \llbracket \text{who } 2. [? t_1] \text{ Sakina see } t_2 \rrbracket^{w, [1 \rightarrow p]}$
 = (by entry for *who* and lambda reduction)
 $\lambda p. \exists x [x \text{ is human in } w \ \& \ \llbracket 2. [? t_1] \text{ Sakina see } t_2 \rrbracket^{w, [1 \rightarrow p]}(x) = 1]$
 = (by Predicate Abstraction and lambda reduction)
 $\lambda p. \exists x [x \text{ is human in } w \ \& \ \llbracket [? t_1] \text{ Sakina see } t_2 \rrbracket^{w, [1 \rightarrow p, 2 \rightarrow x]} = 1]$
 = (by IFA)
 $\lambda p. \exists x [x \text{ is human in } w \ \& \ \llbracket [? t_1] \rrbracket^{w, [1 \rightarrow p, 2 \rightarrow x]}(\lambda w'. \llbracket \text{Sakina see } t_2 \rrbracket^{w', [1 \rightarrow p, 2 \rightarrow x]}) = 1]$

Other vacuous covert operators of this sort are H&K's relative pronouns (and their PRO in ch. 8).

The movement of *who* is evidently overt (pre-SS) movement (in this example). For the movement of the operator, which is not pronounced, we can't tell whether it happens before SS or between SS and LF.

We leave out the vacuous operator OP and just interpret the predicate abstract it creates. $\emptyset$ in the superscript stands for the empty variable assignment, see H&K ch.5

$$\begin{aligned}
&= (\text{by FA and dropping irrelevant superscripts}) \\
&\lambda p. \exists x [x \text{ is human in } w \ \& \ \llbracket ? \rrbracket (\llbracket t_1 \rrbracket^{[1 \rightarrow p, 2 \rightarrow x]})(\lambda w'. \llbracket \text{Sakina see } t_2 \rrbracket^{w', [1 \rightarrow p, 2 \rightarrow x]}) = 1] \\
&= (\text{by Traces rule}) \\
&\lambda p. \exists x [x \text{ is human in } w \ \& \ \llbracket ? \rrbracket (p)(\lambda w'. \llbracket \text{Sakina see } t_2 \rrbracket^{w', [1 \rightarrow p, 2 \rightarrow x]}) = 1] \\
&= (\text{by entry for } ?) \\
&\lambda p. \exists x [x \text{ is human in } w \ \& \ p = \lambda w'. \llbracket \text{Sakina see } t_2 \rrbracket^{w', [1 \rightarrow p, 2 \rightarrow x]}] \\
&= (\text{by FA, entries for } \textit{Sakina}, \textit{see}, \text{Traces Rule}) \\
&\lambda p. \exists x [x \text{ is human in } w \ \& \ p = \lambda w'. \text{Sakina sees } x \text{ in } w']
\end{aligned}$$

This characterizes a set of propositions that contains one proposition per human-in- w : the proposition that that human was seen by Sakina.

This is the ANSPOSS in Hagstrom 2003's terminology.

How about polar and alternative questions? Can we just posit the same operators in the C-head? Let's try.

(11) Did Sakina see Emily?

$$\begin{aligned}
&[_C ? \text{OP}] \text{Sakina see Emily} \\
&\text{OP } 1 \llbracket ? t_1 \rrbracket \text{Sakina see Emily}
\end{aligned}$$

Here we show an LF and computation that includes the vacuous operator. As an exercise, convince yourself that in the case of polar questions, a simpler structure without OP is likewise interpretable, with equivalent results.

(12) Computation for LF of the polar question (11):

$$\begin{aligned}
&\llbracket 1. \llbracket ? t_1 \rrbracket \text{Sakina see Emily} \rrbracket^{w, \emptyset} \\
&= (\text{by Predicate Abstraction}) \\
&\lambda p_{st}. \llbracket \llbracket ? t_1 \rrbracket \text{Sakina see Emily} \rrbracket^{w, [1 \rightarrow p]} \\
&= (\text{by IFA}) \\
&\lambda p_{st}. \llbracket \llbracket ? t_1 \rrbracket \rrbracket^{w, [1 \rightarrow p]} (\lambda w'. \llbracket \text{Sakina see Emily} \rrbracket^{w', [1 \rightarrow p]}) \\
&= (\text{by FA}) \\
&\lambda p_{st}. \llbracket \llbracket ? \rrbracket \rrbracket^{w, [1 \rightarrow p]} (\llbracket t_1 \rrbracket^{w, [1 \rightarrow p]})(\lambda w'. \llbracket \text{Sakina see Emily} \rrbracket^{w', [1 \rightarrow p]}) \\
&= (\text{by Traces Rule and dropping irrelevant assignment superscripts}) \\
&\lambda p_{st}. \llbracket \llbracket ? \rrbracket (p)(\lambda w'. \llbracket \text{Sakina see Emily} \rrbracket^{w'}) \\
&= (\text{by entry for } ? \text{ and lambda reduction}) \\
&\lambda p_{st}. [p = \lambda w'. \llbracket \text{Sakina see Emily} \rrbracket^{w'}] \\
&= (\text{by FA and entries for } \textit{Sakina}, \textit{Emily}, \textit{see}) \\
&\lambda p_{st}. [p = \lambda w'. \text{Sakina see Emily in } w']
\end{aligned}$$

This is the characteristic function of a singleton set containing one proposition. Is this a good result? If we have in mind that our denotations for interrogative clauses should directly correspond to an intuitive notion of “possible answer”, then this is problematic. There is certainly more than one possible answer to a polar question! But we think of the relation between the semantics and the pragmatics of interrogative clauses in a less simple-minded way. A singleton set of a proposition is used to induce a two-way partition of the context set: one cell containing worlds where the proposition is true and another where it isn't. Any relevant response needs to eliminate one of the cells.

To complete the current section, we consider an alternative question:

- (13) Did Sakina see Emily /or did he see Julie?
 DS: $[[[_C ? OP] \text{ Sakina see Emily}] \text{ or } [[[_C ? OP] \text{ Sakina see Julie}]]$
 LF: $OP \ 1 \ [\ [? \ t_1] \text{ Sakina see Emily}] \text{ or } [? \ t_1] \text{ Sakina see Julie}]$

The slashes are intended as a crude representation of the distinctive intonational contour that characterizes the alternative-question reading.

In the alternative question in (13), operator movement must be “across the board” (ATB), with the result that a single binder binds two coindexed traces. This is the only way to obtain an interpretable structure, given the semantic type of *or*, which is $\langle t, \langle t, t \rangle \rangle$.

- (14) Computation for LF of the alternative question in (13):
 $\llbracket 1. [? \ t_1] \text{ Sakina see Emily or } [? \ t_1] \text{ Sakina see Julie} \rrbracket^{w,0}$
 = (by Predicate Abstraction)
 $\lambda p_{st}. \llbracket [? \ t_1] \text{ Sakina see Emily or } [? \ t_1] \text{ Sakina see Julie} \rrbracket^{w,[1 \rightarrow p]}$
 = (by FA twice and entry for *or*)
 $\lambda p_{st}. \llbracket [? \ t_1] \text{ Sakina see Emily} \rrbracket^{w,[1 \rightarrow p]} \vee \llbracket [? \ t_1] \text{ Sakina see Julie} \rrbracket^{w,[1 \rightarrow p]}$
 = (by IFA, FA, entry for *?*, etc — see previous computations)
 $\lambda p_{st}. [p = \lambda w'. \llbracket \text{Sakina see Emily} \rrbracket^{w',[1 \rightarrow p]} \vee p = \lambda w'. \llbracket \text{Sakina see Julie} \rrbracket^{w',[1 \rightarrow p]}]$
 = (by FA and entries for *see* etc.)
 $\lambda p_{st}. [p = \lambda w'. \text{ Sakina see Emily in } w' \vee p = \lambda w'. \text{ Sakina see Julie in } w']$

This is the characteristic function of a set containing two propositions — the same set, in fact, that is denoted by the constituent question *who did Sakina see* if the set of humans happens to be just {Emily, Julie}.

8.2.2 Back to pragmatics: Mapping interrogative denotations to partitions

We can now make good on the promise to provide a bridge between the compositional semantics of interrogatives and their use in updating a context. We gave a draft version earlier in Definition 6, repeated here:

Definition 6 (Update by question (draft)) To ask a question by uttering a sentence ϕ that denotes a set of propositions $\mathcal{P}$ is to propose that the current partitioned context set C be replaced by the new partitioned context set $\text{PART}(\mathcal{P}, \cup C)$.

To finalize the definition, we need to use our semantics to supply a set of propositions to serve as the seed for the partition. This is straightforward in the case of polar and alternative questions, each of which has as its semantic value a set of propositions that is evaluation-world independent. But in the case of *wh*-questions, an issue arises: their semantic value depends non-trivially on the evaluation world.

- (15) $\llbracket \text{Who did Sakina see?} \rrbracket^w$
 $= \lambda p. \exists x: x \text{ is human in } w. p = \lambda w'. \text{ Sakina saw } x \text{ in } w'$

What set of propositions the interrogative denotes in a world w depends on who the humans in that world are. In case this seems esoteric, consider a more substantively restricted wh-phrase:

- (16) $\llbracket \text{Which students called?} \rrbracket^w$
 $= \lambda p. \exists x: x \text{ is a student in } w. p = \lambda w'. \text{ Sakina saw } x \text{ in } w'$

The semantic value of (16) depends on who the students in the evaluation world are. It is easy to imagine circumstances where it is not common ground who the students are. So, how should we update the context with a question like (16)?

Option 1: The world of utterance. We could say that the context gets partitioned based on the set of propositions denoted by the interrogative relative to the world in which the question is asked. However, this would miss the fact that the participants in the conversation do not know which world they are in. The context set is precisely meant to model that they have some common ground on what world they're in but no more: there are always multiple candidates for what the actual world is. Furthermore, it is presumably inescapable that participants make some false presuppositions, in which case the actual world in which the utterance occurs isn't even part of the context set (even though everyone is acting as if the common ground contains the actual world).

So, it seems clear that we need to interpret the interrogative with respect to the worlds in the context set and use the resulting set of propositions to partition the context set. But now we need to face the possibility that the participants in the conversation where (16) is uttered are uncertain about who the (relevant) students are. To set up a minimal test scenario, assume that everyone knows that a is a student but there's uncertainty about whether b is. Now, assume that (16) is uttered against that background. There are worlds in the context set where a and b are the students, and other worlds where only a is a student. Our semantics will deliver two different sets of propositions as the semantic value of the interrogative for the two different kinds of worlds in the context set: for the a, b -worlds, we get the set containing the proposition that a called and the proposition that b called; while for the a -worlds, we get the set that contains only the proposition that a called. How should the interrogative partition the context set?

Option 1 would say something like: To ask a question by uttering a sentence ϕ in world w is to propose that the current partitioned context set C be replaced by the new partitioned context set $\text{PART}(\llbracket \phi \rrbracket^w, \cup C)$.

Option 2: Collecting all the propositions. To obtain the set of propositions that is used to partition the context set, we could collect all the propositions that for *some* world in the context set are in the semantic value of the interrogative:

- (17) The set of propositions that is used to partition a context set c upon the utterance of an interrogative ϕ is

$$\{p: \exists w \in c. p \in \llbracket \phi \rrbracket^w\}$$

For our minimal test scenario, that means that the context set will be partitioned by the the set containing the proposition that a called and the proposition that b called. By our definition of relevance, this means that the answer “ b called” will be relevant. But notice that when the context set is updated by the assertion that b called, there will still be worlds left in the context set where b is not a student (while it is now accepted that b called). This is incorrect: answering that b called when what was asked is *Which students called?* surely commits one to b being a student.

Option 3: Presupposition of consensus. [Stalnaker 1978](#) in his analysis of the speech act of assertion formulated three principles that he views as “essential conditions of rational communication”:

1. A proposition asserted is always true in some but not all of the possible worlds in the context set.
2. Any assertive utterance should express a proposition, relative to each possible world in the context set, and that proposition should have a truth-value in each possible world in the context set.
3. The same proposition is expressed relative to each possible world in the context set.

What’s relevant for us here is the third principle, which ensures that there is common ground on which proposition the speaker is proposing to add to the common ground. We suggest that essentially the same principle applies to the speech act of asking a question: the same set of propositions needs to be expressed by the interrogative relative to each possible world in the context set. This then amounts to the assumption that for each *wh*-phrase that is used, the interlocutors agree on a specific set of individuals as its intended range. This means that in our test scenario, the question *Which students called?* is a pragmatic error: asking the question presupposes that it is common ground who the students are.

Hence, the final version of our definition is as follows:

Definition 7 (Update by question (final)) Asking a question by uttering a sentence ϕ is only felicitous in a partitioned context set C if $\forall w, w' \in \bigcup C: \llbracket \phi \rrbracket^w = \llbracket \phi \rrbracket^{w'}$.

Equivalently, form the big union of all the sets of propositions that are the semantic value of the interrogative in worlds of the context sets:

$$\bigcup_{w \in c} \llbracket \phi \rrbracket^w$$

If felicitous, asking a question by uttering a sentence ϕ is to propose that the current partitioned context set C be replaced by the new partitioned context set $\text{PART}(\llbracket \phi \rrbracket^w, \cup C)$, where w is an arbitrary world in $\cup C$.

8.2.3 Syntax: the role of the $[WH]$ feature

In the theory as it stands so far, the semantic identification of *wh*-words with indefinites (e.g. $\llbracket \text{who} \rrbracket^w = \llbracket \text{somebody} \rrbracket^w$) leads to overgeneration of readings. Consider the minimal pair in (18).

- (18) a. Who did Sakina see?
b. Did Sakina see somebody?

We have shown in Section 8.2.1 how the Karttunen theory derives the intended denotation for (18a) (= (10)). But from the Deep Structure we posited in (10), we could have derived more than one interpretable structure, namely not just (19a) (which we considered in (10) above) but also (19b).

- (19) a. LF-high: OP 1 [who 2 [[? t_1] Sakina see t_2]]
b. LF-low: OP 1 [[? t_1] who 2 [Sakina see t_7]]

Similarly we can derive two interpretable LFs for (18b).

- (20) DS: [C ? OP] Sakina see somebody
a. LF-high: OP 1 [somebody 2 [[? t_1] Sakina see t_2]]
b. LF-low: OP 1 [[? t_1] somebody 2 [Sakina see t_2]]

In all of these potential LFs, the IP-sister of C , as well as the mother-node of C , are semantically of type t . Therefore, both should be suitable adjunction sites for the generalized quantifiers *who* and *somebody* as far as type-compatibility goes. (The mnemonic labels ‘high’ and ‘low’ indicate the difference in the quantifier’s scope.)

But what do these LFs mean? We have already computed (19a) and seen that it expresses an appropriate meaning for (18a). Specifically, if we combine our semantics with our pragmatics, and if the domain of people that *who* ranges over consists of just Emily and Julie, (19a) determines a 4-cell partition. This matches the intuition that the possible fully exhaustive answers to (18a) are that Sakina saw only Emily, that he saw only Julie, that he saw both Emily and Julie, and that he saw neither. But such a 4-cell partition is clearly not what an utterance of (18b) sets up. (18b) is a polar question and elicits answers such as Yes or No, expressing the propositions that Sakina saw somebody and that he saw nobody. So here we want our semantics and pragmatics to determine a 2-cell partition. The LF-low in (20b) does deliver precisely this. (Compute this as an exercise.)

So LF-high in (19a) captures the attested meaning of the wh-question in (18a), and LF-low in (20b) captures the attested meaning of the polar question in (18b). In each case, this is the only attested reading for the English sentence: the wh-question cannot be read as a polar question, and the polar question cannot be read as a wh-question. Our theory so far fails to derive this un-ambiguity. The LF-low in (19b) is semantically equivalent to the LF-low in (20b) and thus represents an unattested polar reading for the wh-question. Similarly, the LF-high in (20a) is equivalent to the LF-high in (19a) and thus represents an unattested wh-question reading for the polar question with *somebody*. We need to amend our theory so that only one of the LFs is generated for each of the sentences.

A standard solution — effectively Karttunen's — is to invoke a syntactic constraint that regulates scopal relations between existential DPs (*who*, *somebody*) and the interrogative complementizer. As our example teaches us, *who* apparently can only scope above ? (more specifically, between ? and its associated *OP*), whereas *somebody* is barred from scoping there and must instead scope below ?. The following stipulation enforces this generalization.

(21) Wh-Licensing Principle:

At LF, a phrase α occupies a specifier position of ? if and only if α has the feature [WH].

(21) relies on appropriate assumptions about which phrases have the feature [WH]. For the time being, assume that certain words such as *who*, *what*, *how* are marked as [WH] in the lexicon. These then will be the only phrases that can be located right above ? in well-formed LFs, and moreover, they cannot be located anywhere else. LF-high for the wh-question in (19a) complies with (21), as does LF-low for the polar question in (20b). LF-low in (19b) is ruled out because it has a phrase marked [WH] in a location other than spec-of-?, and LF-high in (20a) is prohibited because it has a phrase in spec-of-? which lacks [WH].

Exercise 8.1 Consider the multiple wh-question (22).

(22) *Who likes who?*

Propose an LF, say how it is derived in the syntax, compute its semantic interpretation, and then compute the partition it imposes. For simplicity, assume in this last part that the domain of people that *who* ranges over is just $\{s, e\}$, and that the common ground before the question is totally uninformative, i.e. it is a partitioned context set whose union is W (the set of all worlds whatsoever). How many cells does the partition have and what are its cells? Regarding syntax, attend in particular to the satisfaction of the Wh-Licensing Principle.

Exercise 8.2 In written language, a question containing *or*, such as (23), can be ambiguous.

(23) *Did you talk to David or Norvin?*

The questioner may want to know which of the two professors you talked to (“alternative-question reading”) or just whether you talked to at least one of them (“polar-question reading”). Perhaps the alternative reading is more salient out of the blue, but the polar reading can be facilitated by an appropriate context. (E.g. imagine that your squib is on multiple-*wh*-constructions, and you have previously been told that you ought to consult a faculty member who has published on this topic, namely David or Norvin.)

Your task in the exercise is to analyze the two readings of (23), by proposing an appropriate LF for each reading and discussing its syntactic derivation as well as its semantic and pragmatic interpretation. You should assume that English has only one unambiguous word *or*, and its semantic type is $\langle t, \langle t, t \rangle \rangle$. This means that, for both readings, you must posit some amount of elided material in the right disjunct, since *or* can only coordinate constituents of type *t*.

8.3 Embedded questions and question-embedding verbs

A first superficial survey of English verbs that take complement clauses turns up three groups. One group, exemplified by the verb *believe*, consists of verbs that take *that*-clauses but are ungrammatical with an embedded interrogative clause.

White 2021 argues that things are not that simple.

- (24) a. Sakina believes that Becky called.
b. *Sakina believes who called.

Another rather large second group consists of verbs that can take either. This includes *know*, *remember*, and *tell*.

- (25) a. Sakina knows/remembers/told Emily that Becky called.
b. Sakina knows/remembers/told Emily who called.

Third, there are verbs which take interrogative complements but are ungrammatical with *that*-clauses.

- (26) a. *Sakina asked/is wondering that Becky called.
b. Sakina asked/is wondering who called.

We will focus at first on the middle group, which Lahiri 2002 dubbed the class of “responsive” verbs.

8.3.1 Responsive verbs, a type-mismatch, and Dayal’s strategy

Let’s look at the responsive verb *know*. We have at least a preliminary analysis for verbs like *know* in sentences like *John knows that Ann called*. Such verbs take a proposition and a person as their arguments, and their meaning encodes universal quantification over a certain set of possible worlds. Since *know* is a factive verb, it also triggers the presupposition that

its complement is true. For simplicity, we assume here that apart from the factive presupposition, the meaning of *know* is the same as the meaning of *believe*, and so we can write the lexical entry in (27).

$$(27) \quad \llbracket \text{know} \rrbracket^w = \lambda p_{st} : p(w) = 1. \lambda x_e. \forall w' [w' \in \text{DOX}(x, w) \rightarrow p(w') = 1]$$

Here, *DOX* maps an individual *x* and a world *w* into the set of worlds compatible with what *x* believes in *w*.

This entry was designed to work for *know* with a *that*-clause. What would happen if we tried to interpret an LF with an interrogative clause as the sister of *know*? Evidently, we would run into a type-mismatch. On our current analysis of interrogative clauses, their extensions are of type $\langle st, t \rangle$ and their intensions of type $\langle s, \langle st, t \rangle \rangle$. Neither type can compose with the type of *know* in (27) by means of any of our semantic rules.

What shall we do? Older literature on question-embedding made verbs like *know* lexically ambiguous, with distinct (though not unrelated) lexical entries for declarative-taking *know* and interrogative-taking *know*. More recent work has pursued the strategy of positing a single unambiguous verb and readjusting the semantic type of the interrogative complement. Groenendijk & Stokhof 1982 did this first, and another influential version of this approach originates with Dayal 1996. Dayal proposed that the combination of the verb with the Karttunen-denotation of its complement is mediated by an “answer operator”, which maps sets of propositions to propositions. We will follow Dayal’s general strategy in these notes. We will entertain a couple of possible meanings for the answer operator and talk about the empirical considerations that bear on the question which meaning is correct.

The rough intuition to be implemented is that “to know who called” means something like “to know the answer to the question ‘who called?’”. (Paraphrases of this form work for all the verbs in this group, hence Lahiri’s term “responsive verbs”.) The answer to a question is a proposition, hence an object of a suitable semantic type to feed to the meaning of *know* in (27). If the LFs of sentences like (25b) contain an operator that maps a question-denotation to the proposition that’s the answer to that question, we have a solution to the type-mismatch problem.

Since our syntax for interrogative clauses already happens to posit a silent operator at the top edge of the clause (albeit one that we have so far treated as semantically vacuous) we need not actually make the structure more complex. Instead we can assume (following a suggestion by Danny Fox) that our new answer operator appears instead of the previous vacuous one. This means that we base-generate it inside *C* as the sister of *?* and move it up for interpretability, leaving a type- $\langle s, t \rangle$ trace as before. Our LF-structure for a sentence with *know* and an interrogative complement then looks as in (28b).

- (28) a. Sakina knows who called.
 b. Sakina $[_{VP}$ knows $[_{CP}$ ANS 1 [who 2 $[[_C ? t_1] t_2$ called]]]]

We will now focus on the task of proposing a meaning for ANS which not only fixes the type-mismatch, but also yields reasonable truth conditions for the *know*-sentence.

8.3.2 An ANS operator inspired by Karttunen

Our initial proposal for the semantics of ANS effectively follows Karttunen 1977 in terms of the truth conditions it predicts for the *know*-sentence as a whole (although the compositional implementation is different). The ANS operator defined in (29) maps the set of propositions denoted by the interrogative to the single proposition which is the conjunction of all its true elements.

$$(29) \quad \llbracket \text{ANS} \rrbracket^w = \lambda Q_{\langle st, t \rangle}. \lambda w'. \forall p [Q(p) = 1 \ \& \ p(w) = 1 \rightarrow p(w') = 1]$$

$$(29') \quad \llbracket \text{ANS} \rrbracket^w = \lambda Q_{\langle st, t \rangle}. \bigcap \{p \in Q : w \in p\}$$

(the intersection of all members of Q that are true in w)

equivalent formulation (modulo sets vs. characteristic functions)

Let's put this entry to work in a computation for the example sentence.

- (30) a. computation of presupposition:
 Let w be a world. Then
 $\llbracket \text{Sakina knows who called} \rrbracket^w$ is defined
 iff (by FA twice)
 $\llbracket \text{know} \rrbracket^w(\llbracket \text{ANS 1 who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w)(s)$ is defined
 iff (by entry for *know*)
 $\llbracket \text{ANS 1 who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w(w) = 1$
 iff (by FA)
 $\llbracket \text{ANS} \rrbracket^w(\llbracket 1 \text{ who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w)(w) = 1$
 iff by entry for ANS
 $\forall p [\llbracket 1 \text{ who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w(p) = 1 \ \& \ p(w) = 1 \rightarrow p(w) = 1]$
 This is a tautology, so we know that $\llbracket \text{Sakina knows who called} \rrbracket^w$
 is defined for all w .
- b. computation of truth condition:
 Let w be a world. Then
 $\llbracket \text{Sakina knows who called} \rrbracket^w = 1$
 iff (by FA twice)
 $\llbracket \text{know} \rrbracket^w(\llbracket \text{ANS 1 who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w)(s) = 1$
 iff (by entry for *know* and truth of presupposition)
 $\forall w' [w' \in \text{DOX}(s, w) \rightarrow \llbracket \text{ANS 1 who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w(w') = 1]$
- c. We interrupt for an embedded computation:
 $\llbracket \text{ANS 1 who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w(w') = 1$
 small iff (by entry for ANS)
 $\forall p [\llbracket 1 \text{ who 2 ?-}t_1 t_2 \text{ called} \rrbracket^w(p) = 1 \ \& \ p(w) = 1 \rightarrow p(w') = 1]$

Observe that the mother node of *know* is interpreted by plain Functional Application (not by Intensional Functional Application, which would have applied if *know* were taking a declarative complement). This is because the *extension* of the constituent headed by ANS is of type $\langle s, t \rangle$.

iff (by earlier computations)

$$\forall p[\exists x[x \text{ is a human in } w \ \& \ p = \lambda w''. x \text{ called in } w''] \ \& \ p(w) = 1 \rightarrow p(w') = 1]$$

iff (by logic of quantifiers)

$$\forall p \forall x[x \text{ is a human in } w \ \& \ p = \lambda w''. x \text{ called in } w'' \ \& \ p(w) = 1 \rightarrow p(w') = 1]$$

iff (by logic of identity)

$$\forall x[x \text{ is a human in } w \ \& \ [\lambda w''. x \text{ called in } w''](w) = 1 \rightarrow [\lambda w''. x \text{ called in } w''](w') = 1]$$

iff (by λ -reduction)

$$\forall x[x \text{ is a human in } w \ \& \ x \text{ called in } w \rightarrow x \text{ called in } w']$$

d. resuming main computation: ...

iff (by plugging in result of embedded computation)

$$\forall w'[w' \in \text{DOX}(s, w) \rightarrow \forall x[x \text{ is a human in } w \ \& \ x \text{ called in } w \rightarrow x \text{ called in } w']]$$

iff (by logic of quantifiers)

$$\forall x[x \text{ is a human in } w \ \& \ x \text{ called in } w \rightarrow \forall w'[w' \in \text{DOX}(s, w) \rightarrow x \text{ called in } w']]$$

In other words, the sentence *Sakina knows who called* is true in w if and only if, for every person who in fact called in w , Sakina believes (in w) that this person called.

8.3.3 Problems, and an ANS operator inspired by Groenendijk & Stokhof

The essence of ANS in our present analysis is that for any evaluation world w , when applied to the semantic value Q of an interrogative (which is a set of proposition), it gives the proposition that is the conjunction of all the propositions in Q that are true in w . This is well known to make some troublesome predictions. Let's start with the most glaring case, then move to subtler ones, before we diagnose the general problem and proceed to solve it.

The example we analyzed above involved an embedded constituent question. What if we embedded a polar question instead?

- (31) a. Sakina knows whether Emily called.
b. LF: Sakina knows $[_{CP} \text{ANS } 1[[_C? \ t_1] \text{ Emily called }]]$

Recall that our denotation for a polar question is a singleton set. The sister of ANS in (31b) denotes the set whose only member is the proposition that Emily called. If we complete the calculation, we get the following truth condition.

- (32) presupposition of (31b): tautological
 $\llbracket (31b) \rrbracket^w = 1$ iff
 Emily called in $w \rightarrow \forall w'[w' \in \text{DOX}(s, w) \rightarrow \text{Emily called in } w']$

This says that the sentence (31a) is true in w if either one of the following two conditions is met: either (i) Emily did not call in w , or else (ii) she did call in w and Sakina believes in w that she did. This is not satisfactory. What it gets right is that, if Emily called but Sakina is unaware of this, then the sentence is false. But it also predicts that, if Emily didn't call, then the sentence is true no matter what Sakina believes—even if she wrongly believes that she did call.

A gut reaction to this problem is that the culprit is our semantics for polar questions, not our *ANS* operator. This is what Karttunen would have said. Indeed, he gave a different semantics for polar questions and did not have this problem. In our variant of his theory, if we minimally changed the semantics of polar questions so that the sister of *ANS* were to denote the 2-membered set {that Emily called, that Emily didn't call}, the truth conditions would come out correct without any revision to our entry for *ANS*. (Exercise: Convince yourself of this.) This looks like a good way out—at least at first. But when we look at further problem cases, we will come to see it is a move that is neither sufficient nor necessary.

Let's return to the case of the embedded constituent question in (28) and scrutinize the truth conditions we derived in (30) a bit more carefully. Suppose that w is a world in which nobody called. Then the universal quantification we computed in (30b) is trivially true: Whatever Sakina's beliefs in w may be, the material conditional '[x called in $w \rightarrow$ Sakina believes in w that x called]' is true for every x (since the antecedent is always false). So the sentence (28) is predicted to be true, for example, in a world where nobody called but Sakina falsely believes that Emily and Julie called. This does not conform to our intuitions.

Finally, as Groenendijk & Stokhof 1982 forcefully pointed out, even if we only consider worlds in which some people did in fact call, the truth conditions imposed by our current (and Karttunen's) semantics are too lax. Suppose that only Emily called, but Sakina thinks that Emily, Julie, and Delphine all called. Would we say that Sakina knows who called? We'd be reluctant to. But our semantics deems the sentence *Sakina knows who called* to be true in this scenario. After all, Sakina does believe of every person who in fact called (namely, of Emily), that that person called.

This is all that our predicted truth conditions require. If our analysis were right, it simply shouldn't matter how many false beliefs Sakina has about people calling who did not in fact call.

Groenendijk & Stokhof argued that the correct semantics for *Sakina knows who called* is what they dubbed “strongly exhaustive”—i.e., the *know*-sentence is true only when Sakina is fully informed about who called and who didn't. She believes that they called of all the people who did in fact call, and she believes that they didn't call of all the ones who didn't. Can we revise our entry for the answer operator so that it delivers this more stringent truth condition? Yes, here is how.

How might we do that? Perhaps by giving a meaning to *whether*, letting it denote a function that maps a singleton set $\{p\}$ to the set $\{p, \neg p\}$.

The problem we're discussing here was noticed by Karttunen 1977 in a footnote, and he fixed it by complicating his lexical entry for interrogative-taking *know*. He ended up stating the truth condition in the form of a disjunction, with a special clause for the case where the question-denotation only contains false propositions. Heim 1994 showed how to generalize Karttunen's special clause to a general solution for all the problem cases we consider in this section. The solution we will present in these lecture notes is not quite the same as Heim's. See papers by Rullmann & Beck 1998, Beck & Rullmann 1999, Sharvit 2002, and Sharvit & Guerzoni 2003 for discussion and comparison.

(33) strongly exhaustive answer operator:

$$\llbracket \text{ANS} \rrbracket^w = \lambda Q_{\langle st, t \rangle} . \lambda w' . \forall p [p \in Q \rightarrow p(w) = p(w')]$$

(33') $\llbracket \text{ANS} \rrbracket^w = \lambda Q_{\langle st, t \rangle} . \{w' : w' \sim_{Q, W} w\}$

What does this semantics for ANS do? It takes the set of propositions Q denoted by the underlying interrogative and maps the evaluation world to the worlds that agree with it on the truth-value of all propositions in Q . In other words, it maps the evaluation world to its cell-mates relative to Q .

This new semantics solves all the problems that we saw in this section. For embedded polar questions, it delivers the prediction that if Emily didn't call, then *Sakina knows whether Emily called* is only true if Sakina knows that Emily didn't call. For embedded constituent questions, it predicts that if x didn't call, then *Sakina knows who called* is not true unless Sakina knows that x didn't call. As a special case of this latter prediction, we derive that if nobody called, then *Sakina knows who called* is only true if Sakina knows about each person that they didn't call.

Exercise 8.3 *Verify these claims, by doing the computations.*

8.3.4 ANS in matrix questions?

Our latest, strongly exhaustive answer operator bears an obvious logical relation to the pragmatic rule “update by question” (Definition 7) that we posited earlier to link the semantic values of matrix interrogative clauses to the speech acts that they serve to perform. In both cases, we use the set of propositions Q to lump together any evaluation world with those worlds that agree with it on the truth-values of all propositions in Q . In the case of interrogatives embedded under responsive predicates like *know*, this set of worlds that are cell-mates with the evaluation world is fed to the attitude predicate as its prejacent proposition. In the case of matrix questions, we use the set of disjoint but exhaustive cells to partition the context set.

Given this similarity, we may want to consider positing ANS in matrix questions as well, instead of the vacuous covert operator *OP*. This would mean rewriting the “update by question” rule. The idea would be that the pragmatic rule can use the intension of the uttered sentence to construct the cells of the new partition. The recipe, informally, is to apply this intension to each world in the current context set and intersect each result with the current context set.

Definition 8 (Update by question, draft revision) To ask a question by uttering a sentence ϕ is to propose that the current partitioned context set C be replaced by the new partitioned context set

$$\{p : \exists w \in \bigcup C . p = \llbracket \phi \rrbracket^w \cap \bigcup C\}.$$

equivalent formulation, using the definition of “cell-mate” from (4a). Here, capital W in the subscript to the $\sim$ -relation is the set of all possible worlds.

Note that the relevant sentences ϕ are of the form “ANS Q ” and relative to any world w denote the proposition that is true of any world that agrees with w on all the propositions in the set denoted by Q relative to w .

We take each world w in the prior context set in turn. We evaluate the matrix question ϕ ($= \text{ANS } Q$) in w and intersect the resulting proposition with the context set, thus finding those worlds in the context set that agree with w on all the propositions in $\llbracket Q \rrbracket^w$. We collect the resulting set of sets of worlds to serve as the new partitioned context set.

There is a tricky issue here, which is a reprise of what we discussed on page 159ff. Take again a test scenario where everyone knows that a is a student but there's uncertainty about whether b is. And there's maximal uncertainty about who called. Now, assume that (34) is uttered against that background:

(34) Which students called?

There are worlds in the context set where a and b are the students, and other worlds where only a is a student. We saw earlier that in such a context, it is best to rule (34) out as pragmatically infelicitous because the set of propositions denoted by the interrogative varies among the worlds in the context set. This was easy enough to do in the system we were working with at that point. Now, however, we are considering the possibility that what is being uttered is really “ANS which students called”. That structure has different semantic values across the worlds in the context set simply because there are different true answers in those worlds (otherwise why ask the question?). So, we can't enforce a presupposition that the interrogative have the same value (the same strong true answer) across the context set.

In an earlier version of these lecture notes, it was erroneously claimed that we could evade the problem this way.

The set of sets of worlds we would get from Definition 8 in our test scenario would not in fact be a partition of the context set:

- (i) a world w where a called and b is not a student will be lumped with any world where a called and b is a student, no matter whether b called (since the proposition that b called is not in the denotation in w of the underlying interrogative);
- (ii) a world w' where a called and b is a student who called will be lumped only with other worlds where a called and b is a student who called;
- (iii) the two sets generated by w and w' are not disjoint, since they both contain worlds where a called and b is a student who called.

The only way we can see to prevent this situation is to state directly in the “update by question” rule that the update is only felicitous if it results in a partition:

Definition 9 (Update by question, final revision) To ask a question by uttering a sentence ϕ is to propose that the current partitioned context set C be replaced by the new context set

$$\{p: \exists w \in \bigcup C. p = \llbracket \phi \rrbracket^w \cap \bigcup C\}$$

If the resulting set of sets of worlds is not a partition of $\bigcup C$, the utterance is infelicitous.

We conclude this subsection by noting that we have now encountered a kind of presupposition that is not grounded in the denotational semantics of the sentences uttered but emerges from the rules and principles of pragmatics.

8.3.5 Embedding under rogative verbs

Access to the intension of the LF headed by **ANS** also figures plausibly in the semantics of sentences with non-responsive (“rogative”) question-embedders, such as the verbs *ask* and *wonder*. These verbs have lexical meanings that suggest rough paraphrases in which *know* or *tell* is in the scope of another intensional operator, e.g., *ask* = ‘request to be told’, *wonder* = ‘want to know’. Following Groenendijk & Stokhof, let’s hypothesize that these verbs differ in semantic type from responsive verbs. Their (internal) argument is of type $\langle s, \langle s, t \rangle \rangle$ rather than $\langle s, t \rangle$. A concrete entry for *wonder* along these lines is (35).

$$(35) \quad \llbracket \text{wonder} \rrbracket^w = \lambda q_{s,st}. \lambda x_e. \\ \forall w' [w' \in \text{DES}(x, w) \rightarrow \forall w'' [w'' \in \text{DOX}(x, w') \rightarrow q(w')(w'') = 1]]$$

The higher semantic type straightforwardly gives us the prediction that *wonder* cannot combine with a *that*-clause. If we attempted to interpret such an LF, we would encounter a type-mismatch. A *that*-clause has an extension of type t and an intension of type $\langle s, t \rangle$. Whether we used plain FA or IFA, we wouldn’t obtain the type- $\langle s, st \rangle$ -function that *wonder* is looking for. But if we embed an interrogative clause (headed by **ANS**) under *wonder*, we will succeed. The extension of the mother-node of **ANS** is type $\langle s, t \rangle$, and its intension is type $\langle s, st \rangle$. So IFA allows us to interpret the structure.

Exercise 8.4 Convince yourself that, given entry (35), the predicted truth condition for Sakina wonders who called matches the informal paraphrase that we gave in the text.

8.4 Wh-movement with pied-piping and reconstruction

As an example of the phenomenon of so-called pied-piping, we will analyze the sentence *How many cats did you adopt?* Before we get to the point that makes the example interesting for our purposes, we must fill in a rudimentary account of plurals and gradability.

In particular, note that we have derived a condition on any wide-scope restriction on wh-phrases that requires the set characterized by that restriction to be “settled” in the prior context set. It would be interesting to compare this result to ideas about “d-linking” that are found in the literature on questions.

DES maps an individual and a world to those worlds that satisfy the individual’s desires as they are in the given world.

8.4.1 Background on plurals and many

The basic idea in most current semantic treatments of plural DPs is that plural definites and pronouns denote entities in D_e , just like singular definites, pronouns, and proper names. The only difference is that the entities denoted by plurals are more complex (and typically spatially discontinuous). A distinction is made within the domain D_e , between so-called “atoms” or “atomic individuals” (the referents of singular DPs) and “pluralities” or “non-atomic individuals” (the referents of plural DPs). Non-atomic individuals contain atomic individuals as (proper) parts; e.g., if Lucy is one of the players, then $\llbracket \text{Lucy} \rrbracket$ (i.e., Lucy) is an atomic part of $\llbracket \text{the players} \rrbracket$. An atomic individual, on the other hand, has no atomic parts other than itself. (An atom counts as an atomic part of itself.)

Given that D_e contains pluralities along with atoms, predicate extensions of type $\langle e, t \rangle$ are functions that apply to both atoms and pluralities. In the case of common nouns, English has a morphological number distinction which seems to have semantic import:

- (36) a. $\llbracket \text{cat} \rrbracket^w = \lambda x. \text{is a cat in } w$
 b. $\llbracket \text{cats} \rrbracket^w = \lambda x. \text{every atomic part of } x \text{ is a cat in } w$

Being a cat entails being an atomic individual (this is how we agree to understand our metalanguage). Therefore, the denotation of the singular noun as defined in (36a) maps every plurality to 0. The pluralized noun in (36b), on the other hand, maps to 1 those pluralities whose atomic parts are all cats.

Verbs can show morphological number too, but we assume that this is always due to morphological agreement with a number-marked subject, and that the number morphology on the verb is not interpreted itself. As far as semantics is concerned, verbs are “number neutral” and typically can be true indiscriminately of both atoms and pluralities. This is reflected, for example, in a lexical entry like (37).

- (37) $\llbracket \text{meow} \rrbracket^w = \lambda x. \text{every atomic part of } x \text{ meows in } w$

The condition in (37) can be met by both pluralities and atoms. A plurality is mapped to 1 iff all its atomic parts meow, and an atom is mapped to 1 if it itself meows. (Recall that every atom counts as an atomic part of itself.)

We can count the atomic parts of a plural individual. For example, the plural individual composed of Lucy, Kathellen, and Shanice has 3 atomic parts. Let’s have a concise notation for this.

Notice that an entity that $\llbracket \text{cats} \rrbracket^w$ maps to 1 is not *necessarily* a plurality. As interpreted in (36b), the plural noun *cats* is also true of a single cat, because of the fact that an atom is an atomic part of itself. It would be possible to revise (36b) so that it requires x to be non-atomic. But as we will see below, the current formulation actually works better. The reason is, in a nutshell, that “One.” is a perfectly good answer to a *how-many* question.

- (38) Let x be an element of D_e . Then
 $\#(x) :=$ the cardinality of the set $\{y: y \text{ is an atomic part of } x\}$.

With this little bit of plural semantics in place, we can now introduce Hackl 2001's semantics for *many* and an appropriate semantics for interrogative *how* that will go with it. Hackl proposes that *many* is not by itself a quantificational determiner of type $\langle et, \langle et, t \rangle \rangle$. Rather it is looking for an argument which denotes a number, and only after it has been saturated with such an argument, the resulting phrase is a quantificational determiner. So the type of *many* is type $\langle e, \langle et, \langle et, t \rangle \rangle \rangle$ — assuming that numbers are abstract individuals of some kind, hence members of D_e — and its entry is as in (39).

- (39) $\llbracket \text{many} \rrbracket = \lambda n: n \text{ is a number. } \lambda f_{et}. \lambda g_{et}. \exists x[\#(x) = n \ \& \ f(x) = 1 \ \& \ g(x) = 1]$

When building a sentence with *many*, in the simplest case we would fill the first argument slot of *many* with a word that refers to a number. This might be an anaphoric demonstrative pronoun *that*, which in appropriate discourse contexts can refer to a previously mentioned number, say the number 3. Or it could be a numeral word, like *three*, which we take here to have a meaning of type e and be a proper name of the number 3. These options give us interpretable syntactic representations like (40a,b).

- (40) a. $[[\text{that}_7 \text{ many}] \text{ cats}] \text{ meowed}$
 b. $[[\text{three many}] \text{ dogs}] \text{ barked}$

(40a) is straightforwardly pronounced as it stands, whereas for (40b), Hackl assumes that *many* is unpronounced after numeral words, so this structure surfaces as *Three dogs barked*. Let us compute truth-conditions, using (39). Suppose we have a contextually given assignment for (40a) which maps the variable 7 to the number 3, and we evaluate the sentence in the actual world @. Then, by using FA three times to apply $\llbracket \text{many} \rrbracket$ to its three arguments, we obtain the truth-condition in (41).

- (41) $\exists x[\#(x) = 3 \ \& \ \llbracket \text{cats} \rrbracket^@ (x) = 1 \ \& \ \llbracket \text{meow} \rrbracket^@ (x) = 1]$

Now we use our entries for *cats* and *meow*, and this becomes (42).

- (42) $\exists x[\#(x) = 3$
 $\ \& \ \text{every atomic part of } x \text{ is a cat in } @$
 $\ \& \ \text{every atomic part of } x \text{ meows in } @]$

In other words, there is a plurality composed of three meowing cats. Which is loud but correct.

Actually, Hackl assumes (with most of the literature on adjectives and gradability) that there is an additional basic type d (for “degrees”) separate from type e . The number argument of *many* is a special case of a degree argument, and the type for *many* is then $\langle d, \langle et, \langle et, t \rangle \rangle \rangle$.

Hackl's analysis implies that the superficially simplest uses of *many*, as in *Many cats meowed*, are actually more complex at LF: the argument position of *many* is bound by a covert POS (“positive operator”), which is a quantifier over numbers (degrees) and means something like ‘a number (degree) above the contextually specified threshold’.

8.4.2 How many-questions

In a *how-many* question, the argument slot that was saturated by that or three in (40) is instead occupied by the *wh*-word *how*. In Karttunen's theory, this will be an existential quantifier, equivalent to *some number*.

- (43) $\llbracket \text{how} \rrbracket = \lambda f_{et}. \exists n[n \text{ is a number} \ \& \ f(n) = 1]$
(type $\langle et, t \rangle$, a generalized quantifier)

This semantic type is not interpretable in situ as the sister of *many*, and must undergo (covert) movement for interpretability. With this in mind, let's attempt a syntactic derivation for the question *How many cats did you adopt?*

- (44) a. $[_C \ ? \ \text{OP}]$ [you adopted how many cats]
b. $\text{OP} \ 5 \ [_{? \ t_5}]$ [you adopted how many cats]
c. $\text{OP} \ 5$ [how many cats 1 $[_{? \ t_5}]$ you adopted t_1]
d. $\text{OP} \ 5$ [how 2 $[_{t_2}$ many cats 1 $[_{? \ t_5}]$ you adopted t_1]]

We have three movements that derive the LF from the base generated structure in (44a): the movement of the empty OP in (44b), the *wh*-movement of the *wh*-phrase *how many cats* in (44c), and the QR of the quantifier *how* in (44d).

We can check the semantic types to confirm that we have derived an interpretable structure (do this as an exercise). Let's compute what (44d) means. (The details are left as an exercise. Here we conflate sets of propositions with their characteristic functions.)

- (45) $\llbracket (44d) \rrbracket^{\text{Q}} =$
...
 $= \{p: \exists n [n \text{ is a number} \ \& \ \exists x[\#(x) = n \ \& \ \llbracket \text{cats} \rrbracket^{\text{Q}}(x) = 1] \ \& \ p = \lambda w. \llbracket \text{adopt} \rrbracket^w(x)(\text{you})]\}$

With a little bit of Predicate Logic reasoning, we can rewrite this equivalently as follows:

- (46) $\{p: \exists x[\underbrace{\exists n[n \text{ is a number} \ \& \ \#(x) = n]}_{\text{tautology}}] \ \& \ \llbracket \text{cats} \rrbracket^{\text{Q}}(x) = 1 \ \& \ p = \lambda w. \llbracket \text{adopt} \rrbracket^w(x)(\text{you})]\}$

We can now contemplate the underlined part and convince ourselves that this part is a tautology. It just says that *x* has some number or other of atomic parts, which cannot fail to be true. So we might as well drop this conjunct and rewrite (46) as (47).

- (47) $\{p: \exists x[\llbracket \text{cats} \rrbracket^{\text{Q}}(x) = 1 \ \& \ p = \lambda w. \llbracket \text{adopt} \rrbracket^w(x)(\text{you})]\}$

Interestingly now, this is precisely the meaning we would have derived for the question *Which cats did you adopt?* In other words, if (44d) really were the LF (or one of the LFs) for the *how-many*-question *How many cats did you adopt?*, then we would be making the bad prediction that this

The main source for the argument in this section is Arnim von Stechow's paper "Against LF pied-piping" (von Stechow 1996).

For Hackl, it would be type $\langle dt, t \rangle$, a generalized quantifier over degrees.

We are exploiting the equivalence of various scopal arrangements in a formula with existential quantifiers and conjunctions.

The following four are all equivalent.

$$\begin{aligned} & \exists x[Fx \ \& \ \exists y[Rxy \ \& \ Gy]] \\ & \exists x \exists y[Fx \ \& \ Rxy \ \& \ Gy] \\ & \exists y \exists x[Fx \ \& \ Rxy \ \& \ Gy] \\ & \exists y[\exists x[Fx \ \& \ Rxy] \ \& \ Gy] \end{aligned}$$

question is synonymous with (or at least shares a reading with) the *which*-question *Which cats did you adopt?* This would be unfortunate for our theory.

Upon closer inspection of (44d), however, it turns out that our theory does not actually generate this LF. We have neglected to check whether (44d) conforms to the Wh-Licensing Principle, repeated here.

- (48) At LF, a phrase α occupies a specifier position of ? if and only if α has the feature [WH].

In (44d), we have two phrases that are scoped above ?, namely *how* and t_2 *many cats*. Let's say that the positions they occupy both count as specifier positions of ? (similar to what one has to say for multiple questions like *who ate what?* — see Exercise 8.1). Then (48) would require that both of these phrases have the feature [WH]. But only *how* actually does, at least on our current assumption that [WH] is a lexical property possessed by only a small set of words. The other phrase that is scoped above ? in (44d) is t_2 *many cats*, which does not carry the feature [WH]. So the structure (44d) is filtered out by the Wh-Licensing principle as syntactically ill-formed. And this is a good thing, because it means we don't generate the unwelcome reading in (47).

We still have to worry, however, about how we generate the reading that our example actually does have. Is there a second, well-formed, LF for our example? The answer is yes if our syntax allows for “reconstruction”, i.e. some mechanism by which overtly moved phrases can be restored to (one of) their pre-moved positions at LF. In order to satisfy the Wh-Licensing Principle, reconstruction must apply to the phrase t_2 *many cats* (though not, of course, to *how*). Where can this phrase be reconstructed to? Well, if we restored it all the way down to its original base position as the object of *adopt*, it wouldn't be interpretable there, because quantifiers are not interpretable in object positions. But if we can assume that wh-movement proceeds (or at least is allowed to proceed) successive cyclically, and that an object wh-phrase can stop over e.g. at the edge of VP before it moves on to Spec-CP, we can target this intermediate landing site for reconstruction. (Assuming the VP-internal subject hypothesis, VPs are semantically of type *t* and hence quantifiers are interpretable at their edges.) By means of reconstruction, we can thus obtain another LF that is both interpretable and in compliance with the WH-Licensing principle.

- (49) OP 5[how 2[[? t_5] t_2 many cats 1[you adopted t_1]]]

The denotation of (49) (which you should compute as an exercise!) is (50).

- (50) $\{p: \exists n[n \text{ is a number} \ \& \ p = \lambda w. \exists x[\#(x) = n \ \& \llbracket \text{cats} \rrbracket^w(x) = 1 \ \& \llbracket \text{adopt} \rrbracket^w(x)(\text{you})]]\}$

This set contains one proposition per number. It contains the proposition that you adopted 1 cat, the proposition that you adopted 2 cats, the proposition that you adopted 3 cats, etc. This prediction accords well with what we feel are expected answers to the question *how many cats did you adopt?*

8.4.3 More on reconstruction

Earlier in these notes in Section 4.2.2, we briefly introduced a version of the “copy theory of movement”, in which reconstruction can be implemented as deletion of a higher copy. We reproduce the relevant passage here:

This assumes that movement generally proceeds in two separate steps, rather than as a single complex operation as we have assumed so far. Recall that in H&K, it was stipulated that every movement effects the following four changes:

- (i) a phrase α is deleted,
- (ii) an index i is attached to the resulting empty node (making it a so-called trace, which the semantic rule for “Pronouns and Traces” recognizes as a variable),
- (iii) a new copy of α is created somewhere else in the tree (at the “landing site”), and
- (iv) the sister-constituent of this new copy gets another instance of the index i adjoined to it (which the semantic rule of Predicate Abstraction recognizes as a binder index).

If we adopt the Copy Theory, we assume instead that there are three distinct operations:

“*Copy*”: Create a new copy of α somewhere in the tree, attach an index i to the original α , and adjoin another instance of i to the sister of the new copy of α . (= steps (ii), (iii), and (iv) above)

“*Delete Lower Copy*”: Delete the original α . (= step (i) above)

“*Delete Upper Copy*”: Delete the new copy of α and both instances of i .

The Copy operation is part of every movement operation, and can happen anywhere in the syntactic derivation. The Delete operations happen at the end of the LF derivation and at the end of the PF deletion. We have a choice of applying either Delete Lower Copy or Delete Upper Copy to each pair of copies, and we can make this choice independently at LF and at PF. (E.g., we can do Copy in the common part of the derivation and then Delete Lower Copy at LF and Delete Upper Copy at PF.)

When we try to apply this machinery to reconstruction of pied-piped material in *wh*-movement, there is a detail that needs attention: After having created two copies (or more) of *how many cats*, what happens when we “move” *how*? Presumably we create another, higher, copy of *how* and coindex it with the lower copy of *how*. With which lower copy of *how*? With the one in the uppermost copy of *how many cats*? If so, then after deletion of that uppermost copy of *how many cats*, the binder index next to the top copy of *how* will no longer bind any trace and the structure will not be interpretable. What we want is to end up with is an LF in which the top copy of *how* binds a variable in a lower (retained) copy of *how many cats*. There may be various ways to achieve this. Perhaps we can simply choose a derivation in which *how* “moves” out of a lower copy of *how many cats*, not out the highest copy. (Richards 1998’s Principle of Minimal Compliance might explain why this long movement is legitimate and yet relies for its legitimacy on the previous movement of the larger phrase.) Alternatively, we may posit a general principle “Copies must remain copies”, which says that, as long as a structure contains more than one copy of a given phrase, every alteration to one of these copies must identically affect them all. In our case, this would mean that when we create the top copy of *how*, we must coindex it with all its existing copies, in the highest as well as all lower instances of *how many cats*. This principle could hopefully be made to fall out from a suitable formalization of copy theory, perhaps in a multi-dominance framework.